Problem Sheet: Functions 1

Jeffrey Thompson

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Exercise 1. Let f be a function defined by

$$f(x) = \frac{3x+2}{7}.$$

1. Find $ff(x)$, and write your answer in the form

$$ff(x) = \frac{ax+b}{49}$$

   where a, b are numbers.

2. Find $f^{-1}(x)$, and write your answer in the form

$$f^{-1}(x) = \frac{ax+b}{3}$$

   where a, b are numbers.

Exercise 2. Let f be a function defined by

$$f(x) = \frac{3x+2}{6x}, \qquad x \neq 0.$$

1. Find $ff(x)$, simplifying your answer into the form

$$ff(x) = \frac{ax+b}{cx+d}$$

   where a, b, c, d are numbers.

2. State where $ff(x)$ is defined.

3. Find $f^{-1}(x)$ in the form

$$f^{-1}(x) = \frac{1}{ax+b}$$

   where a, b are numbers.

4. State where $f^{-1}(x)$ is defined.

Exercise 3. Let f and g be functions defined by

$$f(x) = \sin(x) + \cos(2x)$$

$$g(x) = \frac{x-1}{2}.$$

1. Find $fg(x)$, simplifying your answer.

2. Let $h(x) = \frac{3x+2}{7}$ and $k(x) = \frac{7x-2}{3}$. By computing $hk(x)$, verify that k is the inverse of h.